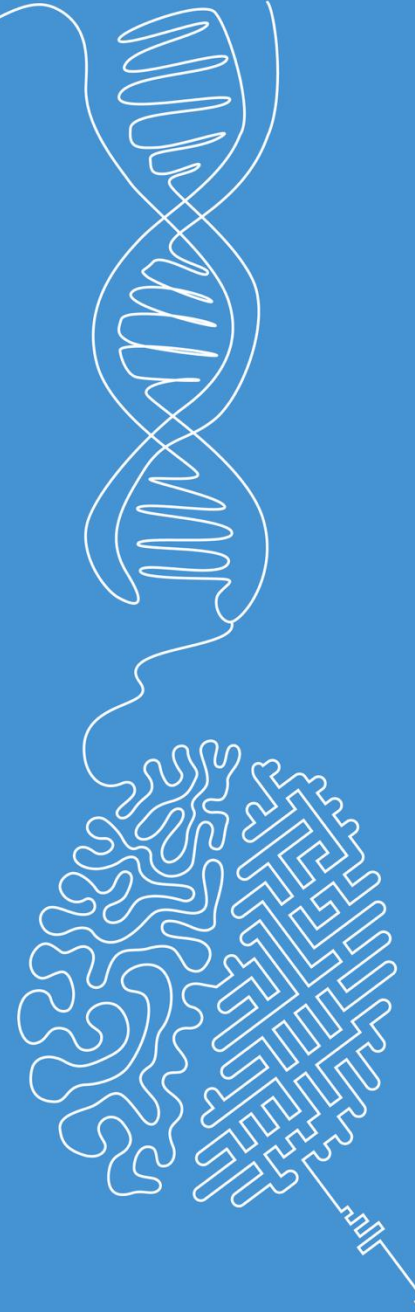


Image segmentation

Image and Signal Processing

Norman Juchler



Classification tasks

Classification in image processing

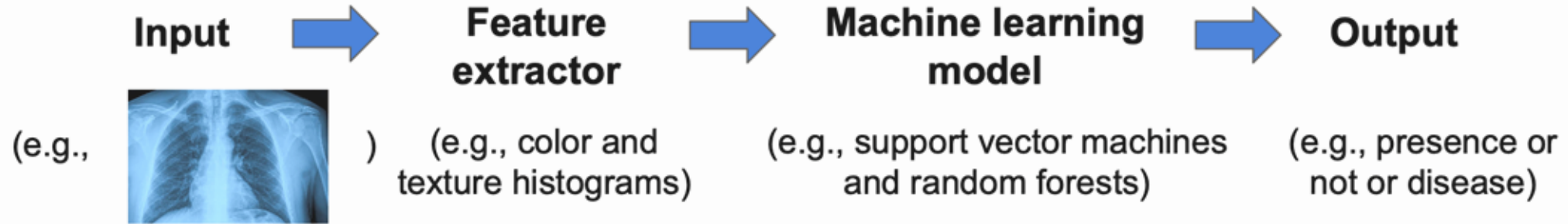
- **Classification task:** Automatically assign labels based on visual characteristics in images, with the goal of identifying or describing specific structures of interest.
- **Levels of classification, illustrated in a health context:**
 - **Subject-level:** Given patient data (including one or more medical images), determine if the patient is healthy or sick.
 - **Image-level:** Decide whether an image indicates the presence of a pathological condition based on its visual characteristics.
 - **Patch-level:** Classify the pathological status of a local region or group of pixels. Example: Are skin lesions benign or malignant?
 - **Pixel-level:** Assign a class label to each pixel. This type of classification is typically referred to as segmentation (see below).



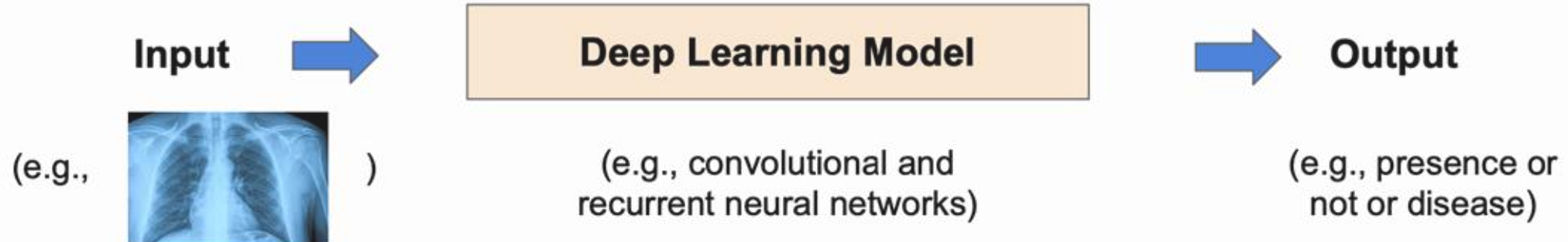
Esteva et al. 2017

How to approach classifications problems?

Traditional machine learning

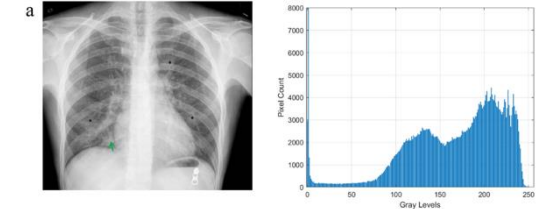


Deep learning

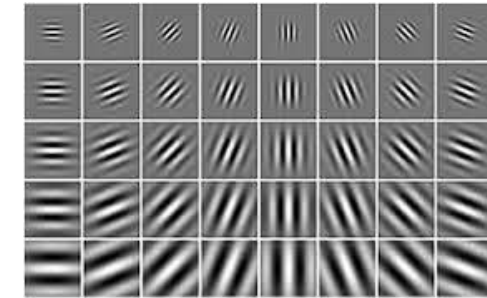


Feature extraction for classification: Traditional approach

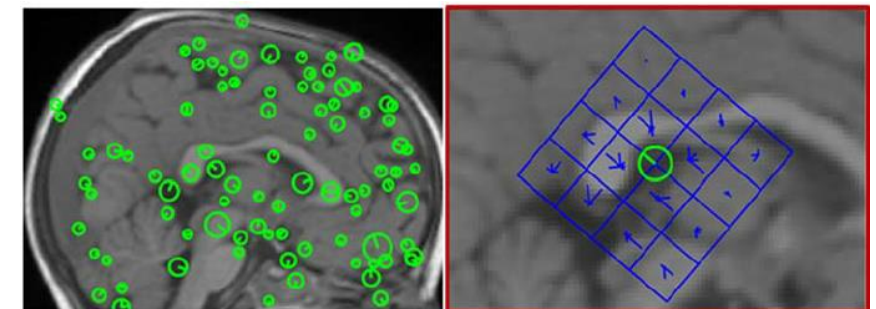
- Types of features (non-exhaustive list):
 - Intensity-based features
 - Histogram-based / statistical features (mean, std, moments, skewness, kurtosis)
 - Gradient-derived features
 - Texture-based features
 - Gray-level co-occurrence matrix features
 - Gabor filter responses
 - Shape-based features
 - Geometric features (area, size ratios, perimeter, etc...)
 - Fourier descriptors
 - Local descriptors
 - SIFT (Scale-Invariant Feature Transform)
 - SURF (Speeded Up Robust Features)
 - Harris corner detector
 - HOG (Histogram of Oriented Gradients)
 - ...



Histogram-based features



Example for Gabor filters

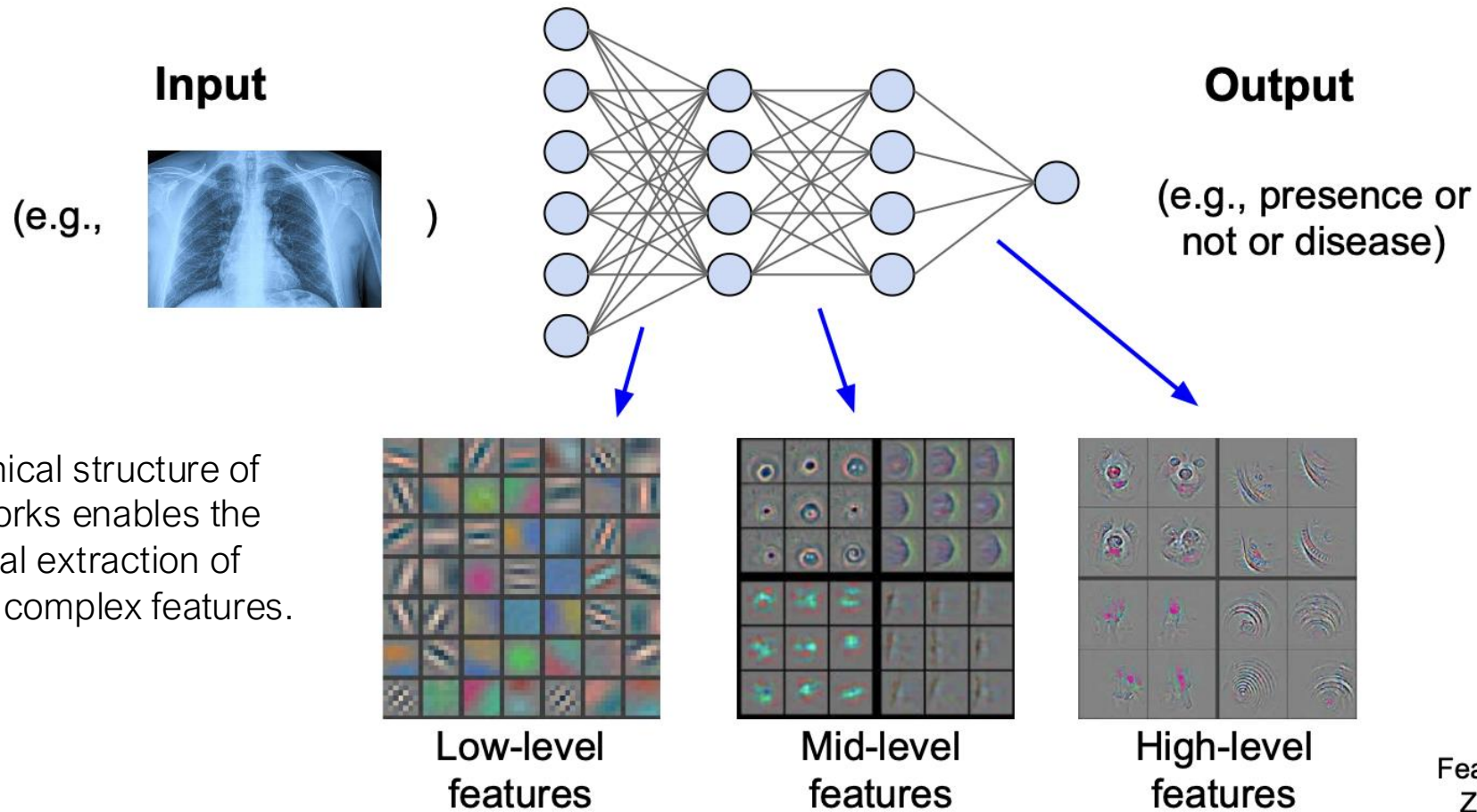


a) SIFT features

b) Appearance matrix

SIFT features (with local descriptor on right side image)

Feature extraction: Deep learning

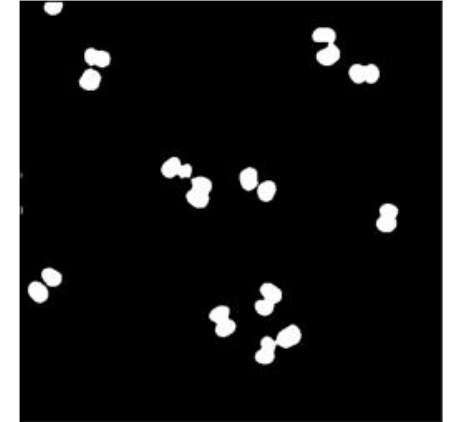
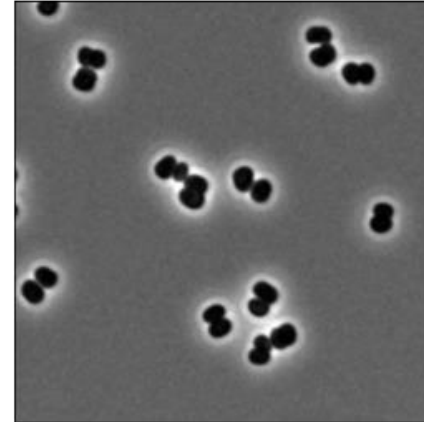


Segmentation

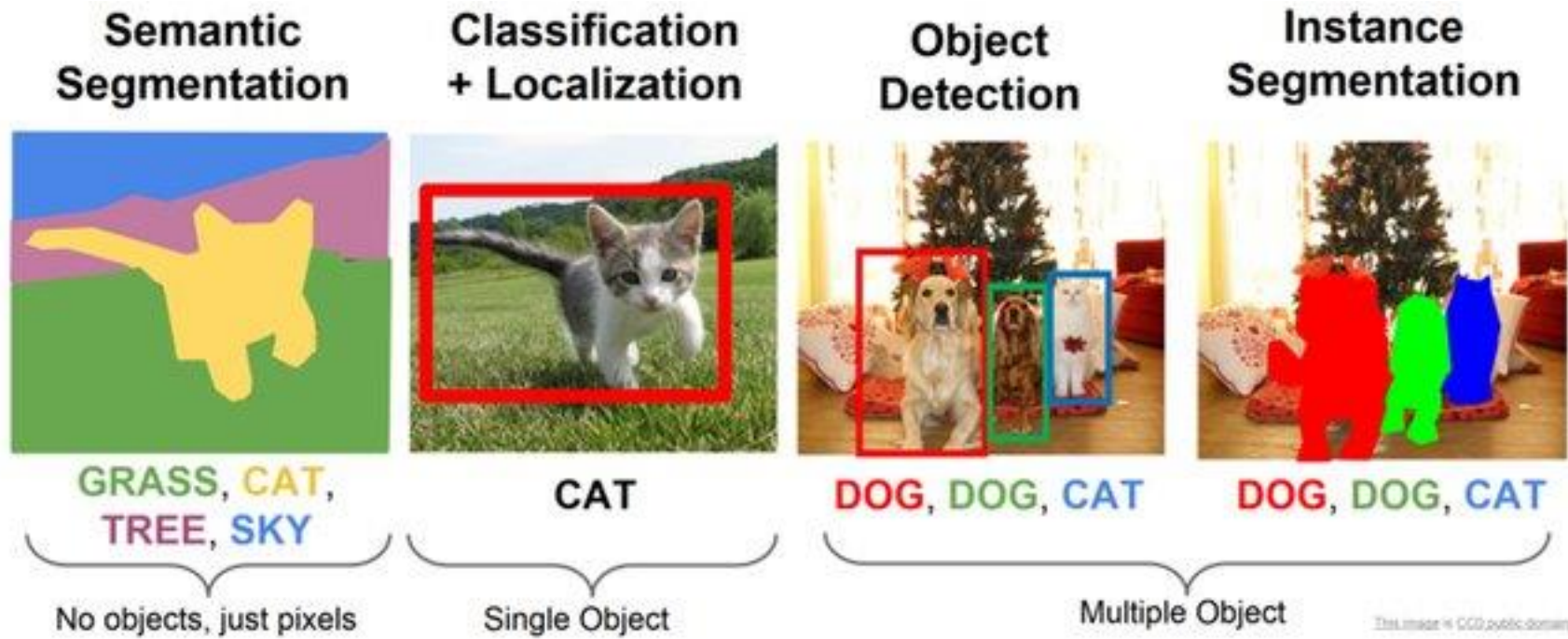
Segmentation

- Image segmentation is the process of partitioning an image into distinct, non-overlapping regions that together cover the entire image.
- The task can be approached in different ways
 - Image processing / image analysis (rule-based, model-driven methods)
 - Machine learning (learning patterns from data)

Focus here: Image processing!



Segmentation vs. classification vs. detection



Segmentation vs. classification vs. detection

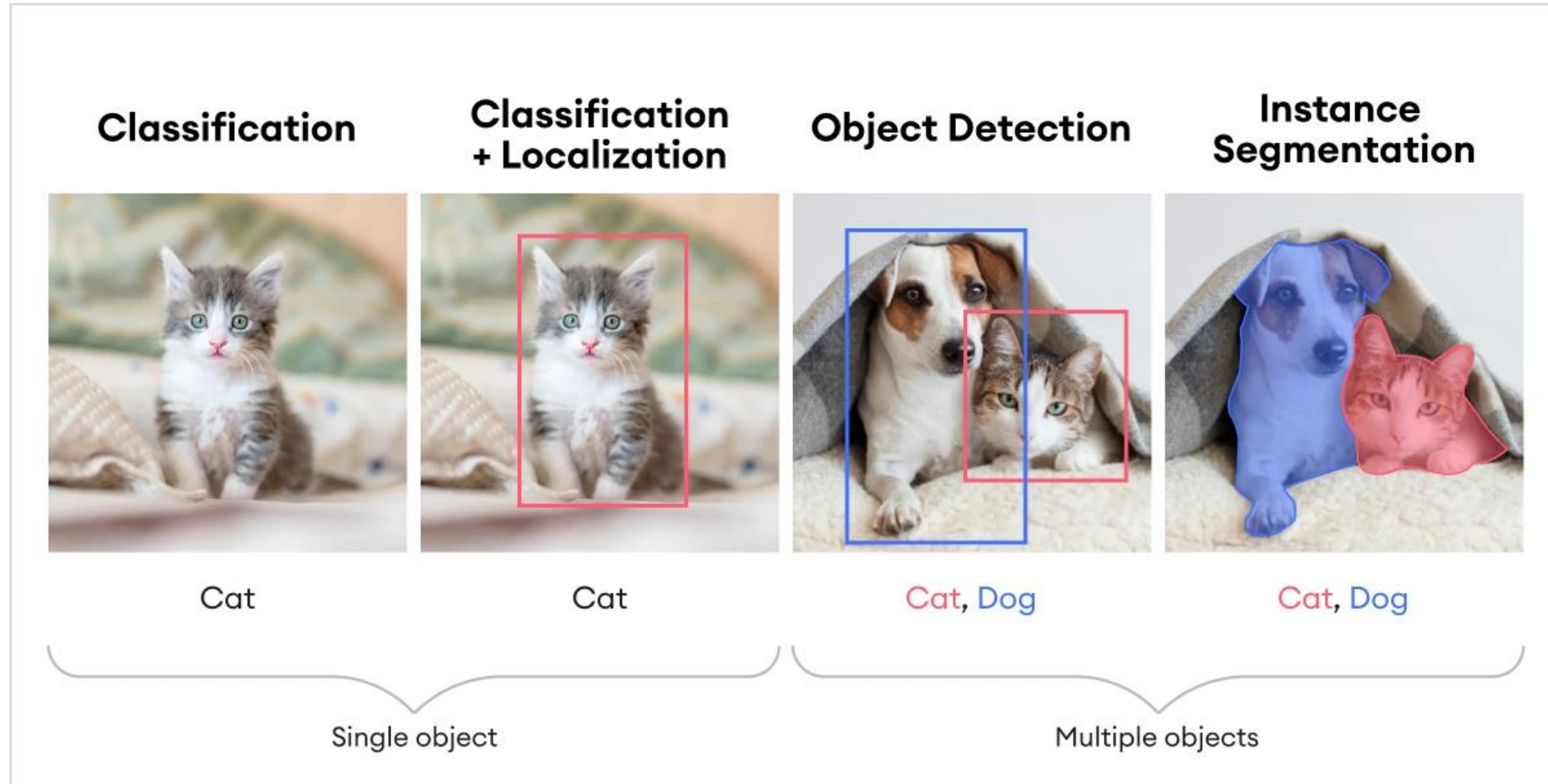


Example for semantic segmentation.
All pixels in the image have a label,
same objects have the same label.



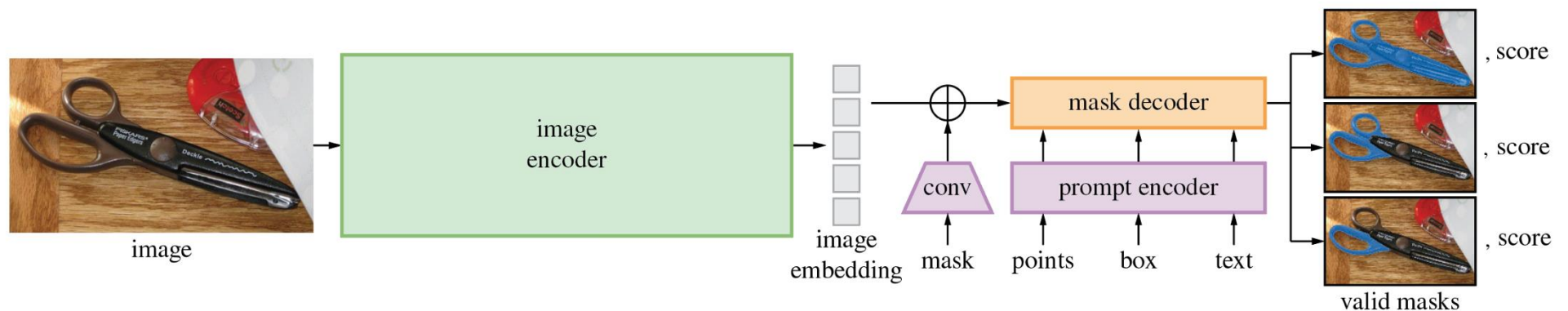
Example for instance segmentation.
Instances of the same object have
different labels (but still, objects of the
same class can be grouped together)

Segmentation vs. classification vs. detection



Demo: Segment anything model (SAM)

- Automatically generates segmentation masks for arbitrary images from simple input prompts, such as points or bounding boxes.
- Demo: <https://aidemos.meta.com>
- Quick technology overview:
 - [Summary by Meta AI](#)
 - [YouTube video \(by No Hype AI\)](#)
- Article (about SAM 1): [Kirillov et al., 2023](#)



Demo: Segment anything model (SAM)

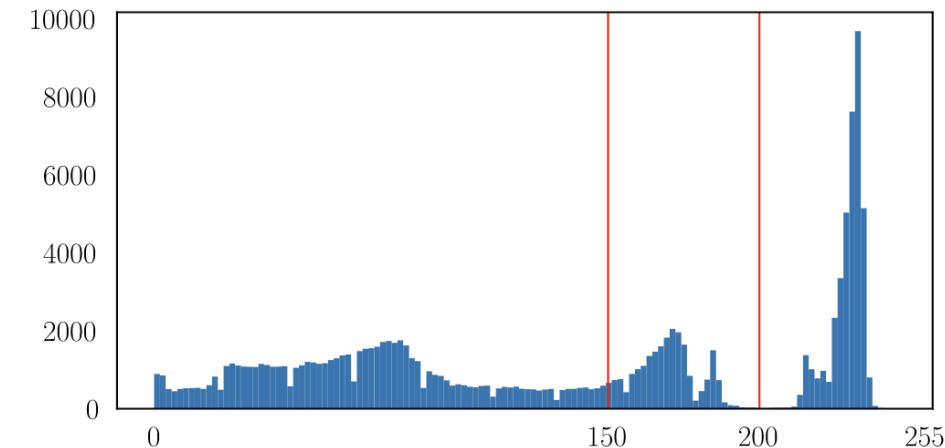
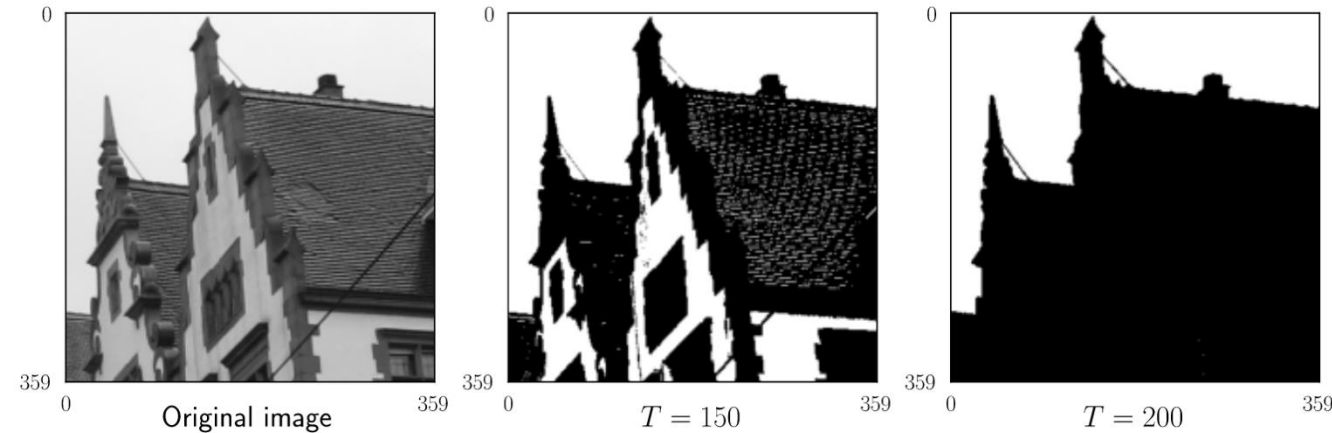
- Deep-learning-based image segmentation method by Meta AI
- Foundation model designed to generalize across domains and tasks
- Generates segmentation masks from simple prompts (points or bounding boxes)
- Trained on >1B masks from 11M images.
- Zero-shot capability: Segment new objects and domains without retraining.
- Open-source, with released model weights for broad integration.



Binary thresholding

- Simplest form of image segmentation
- Partitions an image into foreground and background by comparing pixel intensities to a fixed threshold.
- Mathematical formulation:
 - Input: Grayscale image $I(x, y)$
 - Parameter T (local or global)

$$B(x, y) = \begin{cases} 1, & \text{if } I(x, y) \geq T \\ 0, & \text{if } I(x, y) < T \end{cases}$$



Binary thresholding – Otsu's method

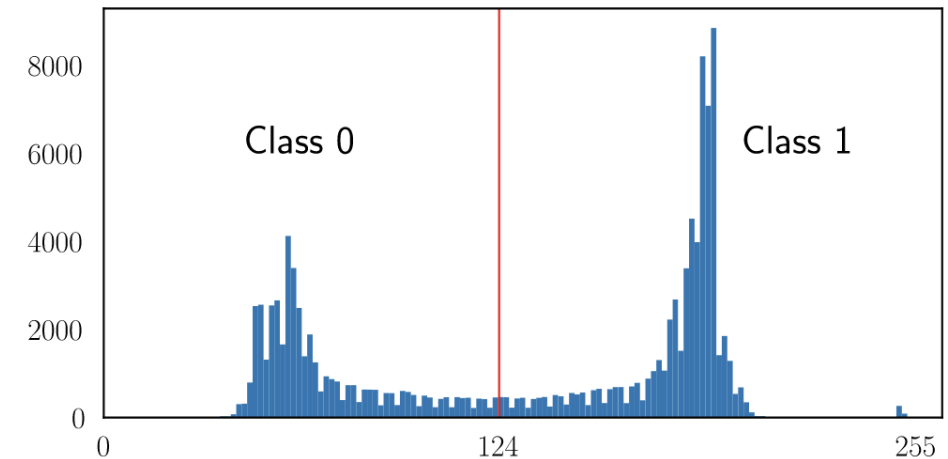
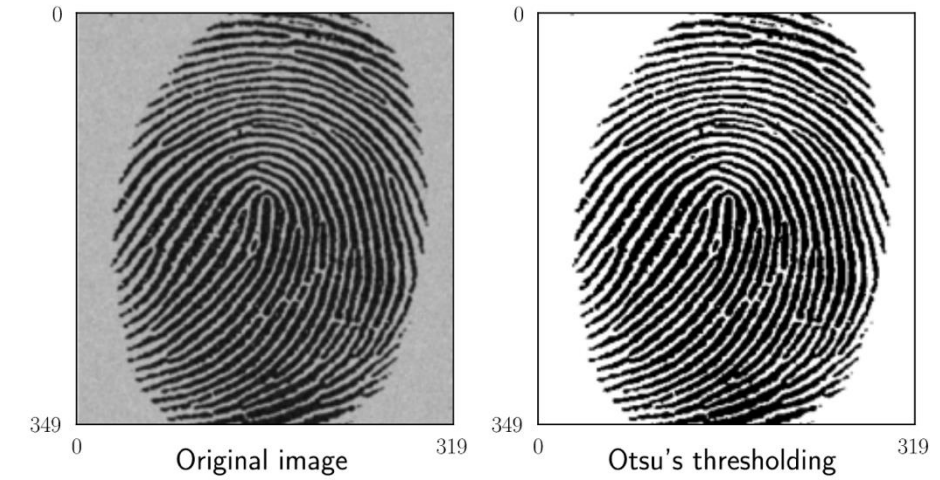
- Mathematical formulation:

- Input: Grayscale image $I(x, y)$
- Parameter T (local or global)

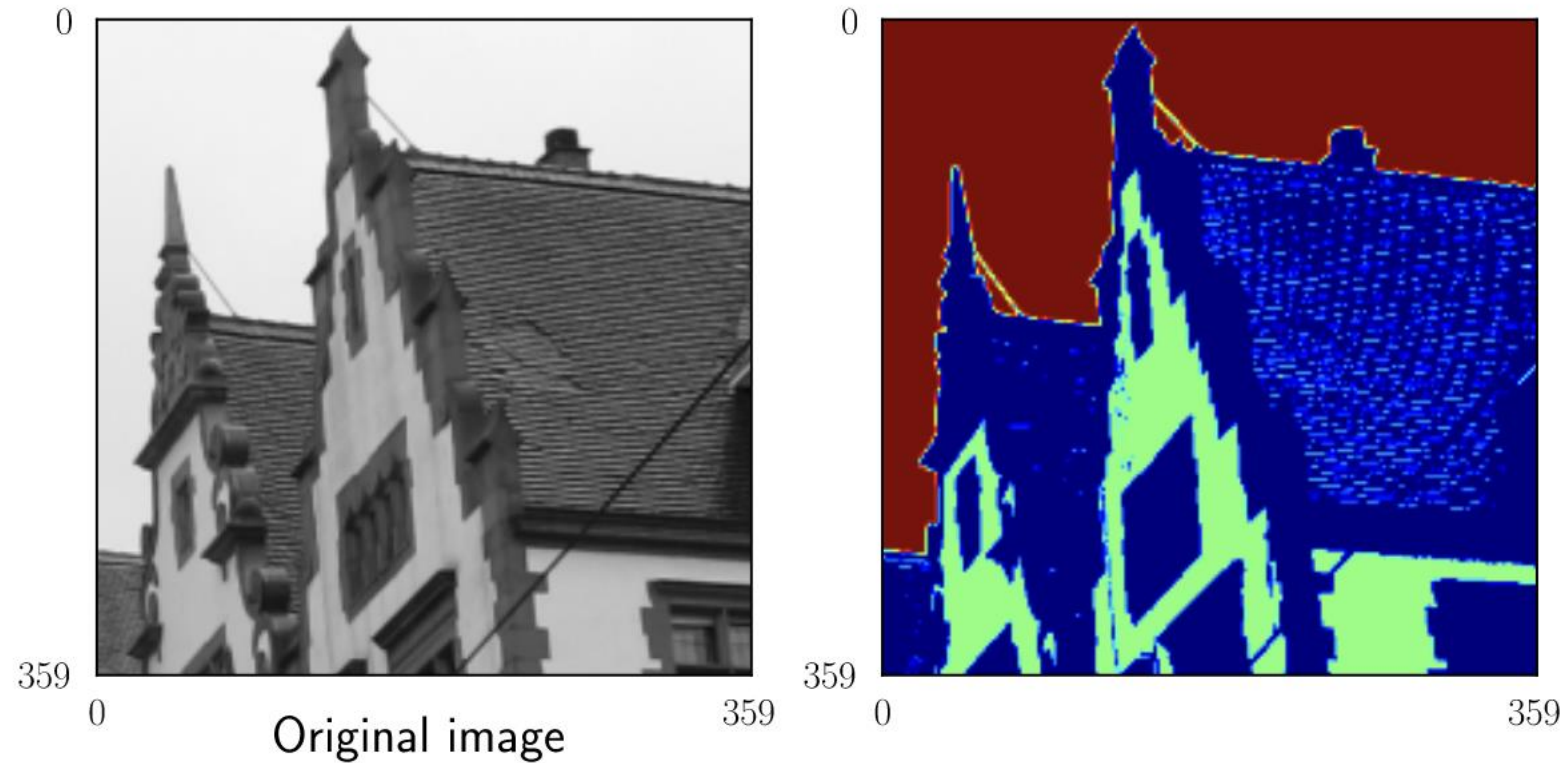
$$B(x, y) = \begin{cases} 1, & \text{if } I(x, y) \geq T \\ 0, & \text{if } I(x, y) < T \end{cases}$$

- Goal: Find the optimal threshold value T that separates image pixels into two classes by **minimizing intra-class variance**.

- Works best if the image has a bimodal histogram

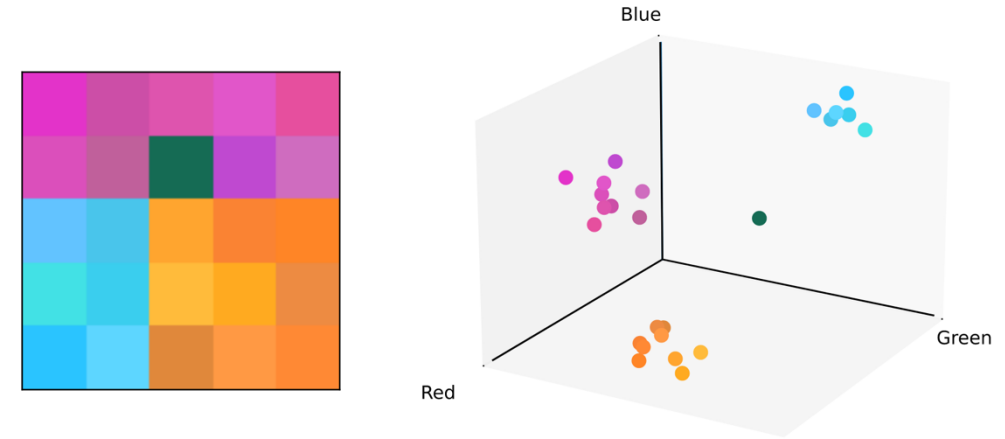


Binary thresholding – Multiple thresholds



Color clustering

- Idea: Segment an image by grouping pixels with similar values / colors into clusters.
- Approach: Pixels are represented in a color space (e.g., RGB, HSV, Lab) and clustered based on color similarity.
- Common method: k-means clustering
- Advantages:
 - Simple and easy to implement
 - Unsupervised (no labeled data required)
- Disadvantages:
 - The number of clusters must be specified *a priori*
 - Sensitive to initialization and parameter choices



k-means clustering

1. Initialize

- Randomly initialize K cluster centroids.

2. Iterate until convergence:

- **Step A (Assignment):** For each data point
 - Compute its distance to each centroid
 - Assign the point to the nearest cluster
- **Step B (Update):** Recompute each centroid as the mean of all points assigned to that cluster

